Arbitrarily Long Increasing Chains of Squares Whose Concatenation Remains a Square

Qichao Wang
Hebei University of Technology, Tianjin, China

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Abstract

We give an elementary, constructive existence theorem for concatenations of perfect squares. Every positive square X^2 can be extended to a larger square by appending a nonzero square having an arbitrarily large prescribed even digit length from an infinite set of lengths. The appended block may be arbitrarily large. Iteration proves that for each $n \geq 1$ there exist strictly increasing positive squares $s_1 < \cdots < s_n$ whose decimal concatenation $s_1 \,\|\, \cdots \,\|\, s_n$ is a square. In particular, OEIS A087354, A391022, and A391189 are defined at every positive index. The argument uses factorization of a difference of squares and the density of the multiplicative group $\{2^u 5^v : u, v \in \mathbb{Z}\}$ in the positive reals. It does not determine the minimum values of the sequences.

1 Background and statement

For positive integers $a_1, \ldots, a_n$, let $a_1 \,\|\, \cdots \,\|\, a_n$ denote their decimal concatenation with no leading zeros in any block. OEIS A087354 records the *least* square representable as a concatenation of n nonzero squares; repetitions are permitted [1]. The more restrictive A391022 requires the blocks to be distinct and strictly increasing, while A391189 minimizes the largest square root in such a representation [2, 3]. The issue addressed here is existence for *every* n, not optimization of any of these minima.

Theorem 1 (Unbounded extension of an arbitrary square)**.** *For each positive integer X and each real bound $M > 0$, there exist positive integers h, Y, Z such that*

$$Z^2 = X^2 10^{2h} + Y^2, \qquad 10^{2h-1} < Y^2 < 10^{2h}, \qquad Y^2 > M. \tag{1}$$

In particular, appending the decimal representation of the nonzero square Y^2 to that of X^2 yields exactly the square Z^2. For each fixed X, there are infinitely many such extensions.

Corollary 1 (All-index existence)**.** *For every integer $n \geq 1$, there exist positive integers*

$$s_1 < s_2 < \cdots < s_n$$

such that every s_i is a square and $s_1 \,\|\, \cdots \,\|\, s_n$ is a square. Consequently, each of A087354(n), A391022(n) and A391189(n) exists and is positive. There are infinitely many such concatenations for each $n \geq 2$.

Proof. Start with $s_1 = 1$ and $X_1 = 1$, so $X_1^2 = s_1$. Suppose $s_1 < \cdots < s_j$ have been chosen and $X_j^2 = s_1 \,\|\, \cdots \,\|\, s_j$. Apply Theorem 1 to $X = X_j$ with $M = s_j$. Set $s_{j+1} = Y^2$ and $X_{j+1} = Z$. The digit-length inequality in (1) ensures that this is precisely concatenation (not addition with spurious leading zeros), and $s_{j+1} > s_j$. Induction proves the claim. At every extension step one can choose arbitrarily large h, producing infinitely many distinct concatenations. Since the set of eligible concatenated squares for each n is a nonempty set of positive integers, its least member exists; similarly, the set of eligible largest square roots has a least member. □

2 The construction

Lemma 1 (Density of decimal-smooth rational ratios)**.** *The multiplicative subgroup*

$$\mathcal{G} = \{2^u 5^v : u, v \in \mathbb{Z}\}$$

is dense in $(0, \infty)$.

Proof. The ratio $\log 2 / \log 5$ is irrational: if it were a/b for nonzero integers a, b, then $2^b = 5^a$, contradicting unique factorization (with signs dealt with by inversion). The fractional parts of integer multiples of an irrational number are dense in $[0, 1]$. Thus $\{u \log 2 + v \log 5 : u, v \in \mathbb{Z}\}$ is dense in $\mathbb{R}$: for a target t, approximate $u(\log 2 / \log 5)$ modulo 1 to $t / \log 5$ modulo 1, and then choose v to adjust the integer part. Exponentiation establishes the assertion. □

Proof of Theorem 1. Fix $X \geq 1$. For $q > 1$ put

$$f_X(q) = \frac{X}{2}(q - q^{-1}).$$

This is a continuous strictly increasing function from 0 to $+\infty$ as q increases from 1 to $+\infty$. Therefore

$$\mathcal{I}_X = \left\{ q > 1 : \frac{1}{\sqrt{10}} < f_X(q) < 1 \right\} \tag{2}$$

is a nonempty open interval. By Lemma 1, it contains a rational number of the special form $q = 2^u 5^v$ with $u, v \in \mathbb{Z}$.

Fix this q, and take any integer $h \geq \max\{|u|, |v|\} + 1$. Define two positive even integers

$$A = X10^h q^{-1} = X2^{h-u}5^{h-v}, \qquad B = X10^h q = X2^{h+u}5^{h+v}. \tag{3}$$

The exponents on the right are nonnegative integers and, because $h \geq |u| + 1$, both A and B are even. Since $q > 1$, $B > A$. Consequently

$$Y = \frac{B - A}{2} \in \mathbb{Z}_{>0}, \qquad Z = \frac{B + A}{2} \in \mathbb{Z}_{>0} \tag{4}$$

are integers, and

$$Z^2 - Y^2 = AB = X^2 10^{2h}. \tag{5}$$

Moreover, (2) gives

$$\frac{Y}{10^h} = f_X(q) \in \left(\frac{1}{\sqrt{10}}, 1\right).$$

Squaring,

$$10^{2h-1} < Y^2 < 10^{2h},$$

so Y^2 has exactly $2h$ decimal digits and is nonzero. The same fixed q works for all larger h, and $Y^2 = 10^{2h} f_X(q)^2 \to \infty$. Taking h arbitrarily large forces $Y^2 > M$ and yields infinitely many extensions. □

3 Examples, variants, and scope

For $X = 1$, the choice $q = 2$, $h = 2$ gives

$$A = 50, \quad B = 200, \quad Y = 75, \quad Z = 125, \qquad \boxed{1 \parallel 5625 = 15625 = 125^2}.$$

An example starting from the strictly increasing two-block square $4 \parallel 9 = 49 = 7^2$ is obtained using $X = 7$, $q = 2^{14}5^{-6} = 16384/15625$, and $h = 15$:

$$Y = 332155892578125, \qquad Z = 7007876107421875,$$

$$\boxed{4 \parallel 9 \parallel 1103275369743709144592285156 25 = 7007876107421875^2.} \tag{6}$$

The third block is the 30-digit square of Y and is larger than 9. All stated equalities can be checked with integer arithmetic. The accompanying script uses rational rather than floating-point inequalities to verify the construction.

The proof works in any integer base $b \geq 2$ that has at least two distinct prime factors. Indeed, choose two such primes $\rho, \sigma \mid b$ and replace q by $\rho^u \sigma^v$, 10^h by b^h, and $1/\sqrt{10}$ by $1/\sqrt{b}$. The subgroup generated by ρ, σ is dense for the same unique-factorization reason; for sufficiently large h the factors $Xb^h q$ and Xb^h/q are integral and of the same parity (both odd if bX is odd; both even otherwise). Thus the same extension and induction work in those bases. We do not claim this argument for prime-power bases.

These are *existence* theorems and may produce very large witnesses. They do not provide the exact minimal values in the three OEIS entries. The related A391022 and A391189 entries list only their early values; neither entry currently states an explicit universal existence theorem. Literature priority beyond the references below remains to be checked independently.

References

[1] OEIS Foundation Inc., *A087354: Smallest square which is the concatenation of n nonzero squares*, https://oeis.org/A087354, accessed October 17, 2026.

[2] OEIS Foundation Inc., *A391022: Minimal square obtained by concatenating n distinct nonzero squares in ascending order*, https://oeis.org/A391022, accessed October 17, 2026.

[3] OEIS Foundation Inc., *A391189: Least upper bound on square roots of n distinct ascending square blocks whose concatenation is square*, https://oeis.org/A391189, accessed October 17, 2026.